VENDORGATE

Vendor Onboarding and Purchase Order Portal

TEST PLAN & EVIDENCE

Case Study No. 88

1. Purpose

This Test Plan defines how the VendorGate prototype will be verified against the functional and non-functional requirements. It covers Boundary Value Analysis, Equivalence Class Testing, Decision Table Testing, functional test cases, traceability, defect tracking, defect metrics, and test evidence.

The test data and results in this academic document are prototype/assumed evidence unless an actual execution result is recorded. Case-study figures are not changed.

2. Testing Scope

Area	Coverage
Vendor Registration	Registration, required information, validation and status management
Purchase Order	PO creation, submission, approval workflow and status changes
Approval Limits	Level assignment using the case-study limits
Vendor Acknowledgement	Acknowledgement of approved Purchase Orders
Audit / Records	Approval records and traceability
UI & Validation	Required fields, invalid input handling and user feedback
Non-Functional	Usability, performance, reliability, security and maintainability checks

3. Test Strategy

Testing uses requirement-based functional testing supported by BVA, equivalence classes and decision tables. Each Functional Requirement (FR) is mapped to at least one test case. Boundary testing is especially important for the Purchase Order approval limits because the case study defines four value ranges.

Test environment: proposed academic prototype using React frontend, Node.js/Express backend and seed/mock data. No persistent database is required for this prototype.

4. Boundary Value Analysis (BVA)

The approval thresholds are tested at and immediately around each boundary. The values below are derived directly from the case-study approval limits.

Boundary	Value Tested	Expected Level
Lower boundary	₹50,000	Level 1
Just above Level 1	₹50,001	Level 2
Upper boundary	₹2,00,000	Level 2
Just above Level 2	₹2,00,001	Level 3
Upper boundary	₹10,00,000	Level 3
Just above Level 3	₹10,00,001	Level 4

5. Equivalence Class Testing

Input Class	Condition	Expected Result
EC1	PO value ≤ ₹50,000	Level 1 approval
EC2	₹50,000 < PO value ≤ ₹2,00,000	Level 2 approval
EC3	₹2,00,000 < PO value ≤ ₹10,00,000	Level 3 approval
EC4	PO value > ₹10,00,000	Level 4 approval
EC5	Missing/invalid mandatory vendor information	Validation message; submission blocked
EC6	Valid vendor information	Registration can proceed

6. Decision Table Testing

Condition / Rule	R1	R2	R3	R4
PO value ≤ ₹50,000	Y	N	N	N
₹50,000 < value ≤ ₹2,00,000	N	Y	N	N
₹2,00,000 < value ≤ ₹10,00,000	N	N	Y	N
PO value > ₹10,00,000	N	N	N	Y
Approval level assigned	L1	L2	L3	L4

7. Functional Test Cases

Test ID	FR	Test Scenario	Test Input	Expected Result
TC-01	FR-01	Vendor registration with valid details	Valid vendor details	Vendor is registered successfully
TC-02	FR-02	Reject incomplete vendor registration	Leave mandatory field blank	Validation message shown and submission blocked
TC-03	FR-03	Manage vendor information	Edit valid vendor details	Updated information is saved in prototype state
TC-04	FR-04	Create Purchase Order	Enter valid PO details	PO is created in Draft state
TC-05	FR-05	Submit PO for approval	Submit valid Draft PO	PO moves to approval workflow
TC-06	FR-06	Assign Level 1 approval	PO value ₹50,000	Level 1 is assigned
TC-07	FR-07	Assign Level 2 approval	PO value ₹50,001	Level 2 is assigned
TC-08	FR-08	Assign Level 3 approval	PO value ₹2,00,001	Level 3 is assigned
TC-09	FR-09	Assign Level 4 approval	PO value ₹10,00,001	Level 4 is assigned
TC-10	FR-10	Approve or reject PO	Approver selects approve/reject	PO status changes accordingly
TC-11	FR-11	Vendor acknowledgement	Approved PO available to vendor	PO becomes Acknowledged after acknowledgement
TC-12	FR-12	Review approval/audit record	Open approval record	Relevant approval details are displayed

8. Requirements Traceability

The following matrix provides one-to-one minimum traceability from the 12 Functional Requirements to test cases.

Requirement	Mapped Test Case	Coverage
FR-01	TC-01	Covered
FR-02	TC-02	Covered
FR-03	TC-03	Covered
FR-04	TC-04	Covered
FR-05	TC-05	Covered

FR-06	TC-06	Covered
FR-07	TC-07	Covered
FR-08	TC-08	Covered
FR-09	TC-09	Covered
FR-10	TC-10	Covered
FR-11	TC-11	Covered
FR-12	TC-12	Covered

9. Defect Log

The defect log is maintained during execution. Since this document is prepared before live prototype execution, no case-study defect count is invented. The table is therefore a ready-to-use execution log.

Defect ID	Test ID	Description	Severity	Status	Evidence
D-001	—	To be recorded during execution	—	Not executed	Screenshot / log
D-002	—	To be recorded during execution	—	Not executed	Screenshot / log
D-003	—	To be recorded during execution	—	Not executed	Screenshot / log

10. Test Metrics

Defect Density = Total confirmed defects ÷ Size of the tested software unit. For this academic prototype, the size unit should be stated before calculating the metric (for example, number of tested requirements or executed test cases).

Defect Removal Efficiency (DRE) = Defects removed before release ÷ (Defects removed before release + defects found after release) × 100. Post-release defects are not yet available at the planning stage, so a final DRE percentage should be entered only after execution.

Metric	Formula	Execution Status
Defect Density	Confirmed defects ÷ selected size unit	Calculate after execution
DRE	Pre-release defects ÷ (Pre-release + post-release defects) × 100	Calculate after execution

11. Test Evidence

Evidence should be captured for each executed test case and inserted against the relevant test ID. Recommended evidence includes screenshots of successful registration, validation messages, PO creation, approval-level assignment, approve/reject actions, vendor acknowledgement and audit/approval records.

Evidence ID	Related Test	Evidence to Capture	Status
E-01	TC-01 to TC-03	Vendor registration screens and validation	To be captured
E-02	TC-04 to TC-05	PO creation and submission	To be captured
E-03	TC-06 to TC-09	Approval-level boundary results	To be captured
E-04	TC-10	Approve/reject result	To be captured
E-05	TC-11	Vendor acknowledgement	To be captured
E-06	TC-12	Approval/audit record	To be captured

12. Test Execution & Closure Criteria

Testing can be considered complete when all 12 functional requirements have at least one executed test case, critical functional defects are resolved or formally accepted, required evidence is captured, and the final defect and test metrics are recorded.

13. Conclusion

The Test Plan provides structured coverage of the VendorGate requirements and directly addresses the case-study testing expectations. BVA and equivalence classes focus on approval-limit boundaries, decision tables verify approval-level logic, and the traceability matrix connects every FR to a test case. Actual execution evidence and final defect metrics should be populated after the prototype is tested.